C U R R I C U L U M

Diploma Program

in

Civil Engineering
Computer Science & Engineering
Electrical & Electronics Engineering
Electrical Engineering
Mechanical Engineering

(Academic Year: 2021-22 onwards)

FIRST YEAR

Arunachal Pradesh State Council for Technical Education

(APSCTE)

(Under the Directorate of Higher and Technical Education)
Government of Arunachal Pradesh

Sample Path: FIRST Semester

Sl. No	Category of Course	Code No.	Course Title	Hours per week			Total contact hrs/ week	Credits
				L	T	P		
1	Basic Science	BS101	Mathematics-I	3	1	0	4	4
2		BS103	Applied Physics-I	2	1	2	5	4
3		BS105	Applied Chemistry	2	1	2	5	4
4	Humanities & Social Science	HS101	Communication Skills in English	2	0	2	4	3
5		HS103	Sports and Yoga/ NCC/ NSS	0	0	2	2	1
6	Engineering Science	ES101	Engineering Graphics	0	0	5	5	2.5
7		ES103	Engineering Workshop Practice	0	0	3	3	1.5
						Total	28	20

Course Code: BS 101
Course Title: Mathematics-I

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
3	1	0	4	4	60	40	100	-	-	-
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.) = 100					

Course Content

Unit-I: Trigonometry

Concept of angles, measurement of angles in degrees, grades and radians and their conversions.
T-Ratios of Allied angles (without proof), Sum, Difference formulae and their applications (without proof). Product formulae Transformation of product to sum, difference and vice versa.
T- Ratios of multiple angles, sub-multiple angles (2A, 3A, A/2).
Graphs of $\sin x$, $\cos x$, $\tan x$ and e^x

Unit-II: Differential Calculus

Definition of function; Concept of limits.
Four standard limits $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)$, $\lim_{x\to a}\left(\frac{a^x-1}{x}\right)$ and $\lim_{x\to a}(1+x)^{\frac{1}{x}}$
Differentiation by definition of x^n, $\sin x$, $\cos x$, $\tan x$, e^x and $\log_a x$
Differentiation of sum, product and quotient of functions. Differentiation of function of a function.
Differentiation of trigonometric and inverse trigonometric functions, Logarithmic differentiation, Exponential functions.

Unit - III: Algebra

Complex Numbers: Definition, real and imaginary parts of a Complex number, polar and Cartesian, representation of a complex number and its conversion from one form to other, conjugate of a complex number, modulus and amplitude of a complex number Addition, Subtraction, Multiplication and Division of a complex number. De-movier's theorem, its application.

Partial fractions: Definition of polynomial fraction proper & improper fractions and definition of partial fractions. To resolve proper fraction into partial fraction with denominator containing non-repeated linear factors, repeated linear factors and irreducible non-repeated quadratic factors. To resolve improper fraction into partial fraction.

Permutations and Combinations: Value of ${}^nP_r$ and ${}^nC_r$

Binomial theorem: Binomial theorem (without proof) for positive integral index (expansion and general form); binomial theorem for any index (expansion without proof) first and second binomial approximation with applications to engineering problems.

Suggested Learning Resources:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 40th Edition, 2007.
2. G. B. Thomas, R. L. Finney, Calculus and Analytic Geometry, Addison Wesley, 9th Edition, 1995.
3. Reena Garg, Engineering Mathematics, Khanna Publishing House, New Delhi (Revised Ed. 2018)
4. V. Sundaram, R. Balasubramanian, K.A. Lakshminarayanan, Engineering Mathematics,
5. Reena Garg & Chandrika Prasad, Advanced Engineering Mathematics, Khanna Publishing House, New Delhi

Course Code: BS 103
Course Title: Applied Physics-I

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
2	1	2	5	4	60	40	100	40	60	100
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.) = 200					

Course Content

Unit-I: **Physical world, Units and Measurements**

Physical quantities; fundamental and derived, Units and systems of units (FPS, CGS and SI units), Dimensions and dimensional formulae of physical quantities, Principle of homogeneity of dimensions, Dimensional equations and their applications (conversion from one system of units to other, checking of dimensional equations and derivation of simple equations), Limitations of dimensional analysis.
Measurements: Need, measuring instruments, least count, types of measurement (direct, indirect), Errors in measurements (systematic and random), absolute error, relative error, error propagation, error estimation and significant figures.

Unit-II: Force and Motion

Scalar and Vector quantities – examples, representation of vector, types of vectors. ***(Short Discussion)***
Force, Momentum, Statement and derivation of conservation of linear momentum, its applications such as recoil of gun, rockets, Impulse and its applications.
Circular motion, definition of angular displacement, angular velocity, angular acceleration, frequency, time period, Relation between linear and angular velocity, linear acceleration and angular acceleration (related numerical), Centripetal and Centrifugal forces with live examples, Expression and applications such as banking of roads and bending of cyclist.

Unit-III: Work, Power and Energy

Work: Concept and units, examples of zero work, positive work and negative work.
Friction: concept, types, laws of limiting friction, coefficient of friction, reducing friction and its engineering applications, Work done in moving an object on horizontal and inclined plane for rough and plane surfaces and related applications.
Energy and its units, kinetic energy, gravitational potential energy with examples and derivations, mechanical energy, conservation of mechanical energy for freely falling bodies, transformation of energy (examples).
Power and its units, power and work relationship, calculation of power (numerical problems).

Unit-IV: **Rotational Motion**

Translational and rotational motions with examples, Definition of torque and angular momentum and their examples, Conservation of angular momentum (quantitative) and its applications.
Moment of inertia and its physical significance, radius of gyration for rigid body, Theorems of parallel and perpendicular axes (statements only), Moment of inertia of rod, disc, ring and sphere (hollow and solid); (Formulae only).

Unit-V: **Properties of Matter**

Elasticity: definition of stress and strain, moduli of elasticity, Hooke’s law, significance of stress-strain curve.
Pressure: definition, units, atmospheric pressure, gauge pressure, absolute pressure, Fortin’s Barometer and its applications.

Surface tension: concept, units, cohesive and adhesive forces, angle of contact, Ascent Formula (No derivation), applications of surface tension, effect of temperature and impurity on surface tension.
Viscosity and coefficient of viscosity: Terminal velocity, Stoke's law and effect of temperature on viscosity, application in hydraulic systems.
Hydrodynamics: Fluid motion, stream line and turbulent flow, Reynold's number Equation of continuity, Bernoulli's Theorem (only formula and numerical) and its applications.

Unit-VI: Heat and Thermometry
Concept of heat and temperature, modes of heat transfer (conduction, convection and radiation with examples), specific heats, scales of temperature and their relationship, Types of Thermometer (Mercury thermometer, Bimetallic thermometer, Platinum resistance thermometer, Pyrometer) and their uses.
Expansion of solids, liquids and gases, coefficient of linear, surface and cubical expansions and relation amongst them, Co-efficient of thermal conductivity, engineering applications.

List of Practical's/Activities

1. To measure length, radius of a given cylinder, a test tube and a beaker using a Vernier caliper and find volume of each object.
2. To determine diameter of a wire, a solid ball and thickness of cardboard using a screw gauge.
3. To determine radius of curvature of a convex and a concave mirror/surface using a spher- ometer.
4. To verify triangle and parallelogram law of forces.
5. To find the co-efficient of friction between wood and glass using a horizontal board.
6. To determine force constant of a spring using Hook's Law.
7. To verify law of conservation of mechanical energy (PE to KE).
8. To find the moment of inertia of a flywheel.
9. To find the viscosity of a given liquid (Glycerin) by Stoke's law.
10. To find the coefficient of linear expansion of the material of a rod.
11. To determine atmospheric pressure at a place using Fortin's barometer.
12. To measure room temperature and temperature of a hot bath using mercury thermometer and convert it into different scales.

NB: From the List of Practical Experiments in Applied Physics I, TWO from the following list may be kept as Demonstrative Experiments. In that situation, Minimum number of EIGHT practical experiments are to be performed by the students.

8. To find the moment of inertia of a flywheel.
9. To find the viscosity of a given liquid (Glycerin) by Stoke's law.
10. To find the coefficient of linear expansion of the material of a rod.

Suggested Learning Resources:

1. Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi
2. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi.
3. Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi
4. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi
5. Engineering Physics by DK Bhhatacharya & Poonam Tandan; Oxford University Press, New Delhi.

Course Code: BS 105
Course Title: Applied Chemistry

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
2	1	2	5	4	60	40	100	40	60	100
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.) = 200					

Course Content

Unit-I: Atomic Structure, Chemical Bonding and Solutions

Rutherford model of atom, Bohr's theory (expression of energy and radius to be omitted), Shapes of s,p and d orbitals, Pauli's exclusion principle, Hund's rule of maximum multiplicity Aufbau rule, electronic configuration.
Concept of chemical bonding – cause of chemical bonding, types of bonds: ionic bonding
(NaCl example), covalent bond (H_2, F_2, HF hybridization in $BeCl_2$, BF_3, CH_4, NH_3, HO_2) and anomalous properties of NH_3, HO_2 due to hydrogen bonding, and metallic bonding.
Solution – idea of solute, solvent and solution, methods to express the concentration of solution- molarity (*M* = mole per liter), ppm, mass percentage, volume percentage and mole fraction.

Unit-II: Water

Graphical presentation of water distribution on Earth (pie or bar diagram). Classification of soft and hard water based on soap test, salts causing water hardness, unit of hardness and simple numerical on water hardness.
Cause of poor lathering of soap in hard water, problems caused by the use of hard water in boiler (scale and sludge, foaming and priming, corrosion etc.)
i). Water softening techniques – soda lime process and zeolite process.
ii). Municipal water treatment (in brief only) – sedimentation, coagulation, filtration, sterilization.
Water for human consumption for drinking and cooking purposes from any water sources and enlist Indian standard specification of drinking water (collect data and understand standards).

Unit-III: Engineering Materials

Natural occurrence of metals – minerals, ores of iron, aluminium and copper, gangue (matrix), flux, slag, metallurgy – brief account of general principles of metallurgy.
Extraction of - iron from haematite ore using blast furnace, along with reactions. Alloys – definition, purposes of alloying, ferrous alloys and non-ferrous with suitable examples, properties and applications.
General chemical composition, composition-based applications (elementary idea only details omitted):
Port land cement and hardening, Glasses Refractory and Composite materials.
Polymers – monomer, homo and co polymers, degree of polymerization, simple reactions involved in preparation and their application of thermoplastics and thermosetting plastics (using PVC, PS, PTFE,
nylon-6, nylon-66, Bakelite only), rubber and vulcanization of rubber.

Unit-IV: Chemistry of Fuels and Lubricants

Definition of fuel and combustion of fuel, classification of fuels, calorific values (HCV and LCV), calculation of HCV and LCV using Dulong's formula.
Proximate analysis of coal solid fuel petrol and diesel - fuel rating (octane and cetane numbers),
Chemical composition, calorific values and applications of LPG, CNG, water gas, coal gas, producer gas and biogas.
Lubrication – function and characteristic properties of good lubricant, classification with examples,

Unit-V: Electro Chemistry

Electronic concept of oxidation, reduction and redox reactions.
Definition of terms: electrolytes, non-electrolytes with suitable examples, Faradays laws of electrolysis and simple numerical problems.
Industrial Application of Electrolysis: Electrometallurgy, Electroplating, Electrolytic refining.

Brief Idea of Primary cells – dry cell, and Secondary cell - commercially used lead storage battery.
Introduction to Corrosion of metals: Definition, types of corrosion (chemical and electrochemical), H_2 liberation and O_2 absorption mechanism of electrochemical corrosion, factors affecting rate of corrosion.
External corrosion preventive measures: Galvanization and Tinning

List of Practical's/Activities

A] Volumetric and Gravimetric analysis:

1. Preparation of standard solution of oxalic acid or potassium permanganate.
2. To determine strength of given sodium hydroxide solution by titrating against standard oxalic acid solution using phenolphthalein indicator.
3. Standardization of KM_nO_4 solution using standard oxalic acid and determine the percentage of iron present in given Hematite ore by KM_nO_4 solution.
4. Iodometric estimation of copper in the copper pyrite ore.
5. Volumetric estimation of total acid number (TAN) of given oil.
6. Volumetric estimation of a) Total hardness of given water sample using standard EDTA solution.
b) Alkalinity of given water sample using 0.01M sulphuric acid
7. Proximate analysis of coal
    a) Gravimetric estimation moisture in given coal sample
    b) Gravimetric estimation ash in given coal sample

B] Instrumental analysis

8. Determine the conductivity of given water sample.
9. Determination of the Iron content in given cement sample using colorimeter.
10. Determination of calorific value of solid or liquid fuel using bomb calorimeter.
11. Determination of viscosity of lubricating oil using Redwood viscometer.
12. Determination of flash and fire point of lubricating oil using Able's flash point apparatus.
13. To verify the first law of electrolysis of copper sulfate using copper electrode.
14. Construction and measurement of emf of elector chemical cell (Daniel cell).
15. To study the effect of dissimilar metal combination.

Suggested Learning Resources:

Text Book of Chemistry for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi, 2017-18.
2) Agarwal, & Shikha, Engineering Chemistry, Cambridge University Press; New Delhi, 2015.
3) C.N. R. Rao, Understanding Chemistry, Universities Press (India) Pvt. Ltd., 2011.
4) Dara, S. S. & Dr. S.S. Umare, Engineering Chemistry, S. Chand. Publication, New Delhi, New Delhi, 2015.
5) Jain & Jain, Engineering Chemistry, Dhanpat Rai and Sons; New Delhi, 2015.

Course Code: HS 101
Course Title: Communication Skills in English

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
2	0	2	4	3	60	40	100	40	60	100
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.): 200					

Course Content

Unit-I: Communication: Theory and Practice
Basics of communication: Introduction, meaning and definition, process of communication etc.
Types of communication: formal and informal, verbal, non-verbal and written Barriers to effective communication.
7 Cs for effective communication (considerate, concrete, concise, clear, complete, correct, courteous).
Art of Effective communication.

- Choosing words
- Voice
- Modulation
- Clarity
- Time
- Simplification of words

Technical Communication.

Unit-II: Soft Skills for Professional Excellence
Introduction: Soft Skills and Hard Skills.
Importance of soft skills.
Life skills: Self-awareness & Self-analysis, adaptability, resilience, emotional intelligence & empathy etc.
Applying soft skills across cultures.
Case Studies ***(Not to limit to 1 or 2 case studies but should give no. of case studies on life skills as provided in the model curriculum.)***

Unit-III: Reading Comprehension Comprehension, vocabulary enhancement and grammar exercises based on reading of the following texts:

Section-1
Malgudi Days: R.K. Narayan (chapters not to be omitted, a few may be given for reading & comprehension assignments)
“The Room on Roof” by Ruskin Bond
“The Gift of the Magi” by O. Henry
“Uncle Podger Hangs a Picture” by Jerome K. Jerome

Section-2
“Night of the Scorpion” by Nissim Ezekiel
“Stopping by Woods on a Snowy Evening” by Robert Frost
“Where the Mind is Without Fear” by Rabindranath Tagore
“Ode to Tomatoes” by Pablo Neruda

Unit-IV: Professional Writing
The art of précis writing,
Letters: business and personnel,
Drafting e-mail, notices, minutes of a meeting etc.
Filling-up different forms such as banks and on-line forms for placement etc.

Unit-V: Vocabulary and Grammar
Vocabulary of commonly used words
Glossary of administrative terms (English and Hindi)
One-word substitution, Idioms and phrases etc.
Parts of speech, active and passive voice, tenses etc., Punctuation

List of Practical's / Activities

Unit-I: Listening Skills
Listening Process and Practice: Introduction to recorded lectures, poems, interviews and speeches, listening tests.

Unit-II: Introduction to Phonetics
Sounds: consonant, vowel, diphthongs, etc. transcription of words (IPA), weak forms, syllable division, word stress, intonation, voice etc.

Unit-III: Speaking Skills
Standard and formal speech: Group discussion, oral presentations, public speaking, business presentations etc. Conversation practice and role playing, mock interviews etc.

Unit-IV: Building vocabulary
Etymological study of words and construction of words, phrasal verbs, foreign phrases, idioms and phrases. Jargon/ Register related to organizational set up, word exercises and word games to enhance self-expression and vocabulary of participants.

Suggested Learning Resources:

1. J. D. O'Connor. *Better English Pronunciation.* Cambridge: Cambridge University Press, 1980.
2. Lindley Murray. *An English Grammar: Comprehending Principles and Rules.* London: Wilson and Sons, 1908.
3. Kulbhushan Kumar, Effective Communication Skills, Khanna Publishing House, New Delhi (Revised Edition 2018)
4. Margaret M. Maison. *Examine your English.* Orient Longman: New Delhi, 1964.
5. M. Ashraf Rizvi. *Effective Technical Communication*. Mc-Graw Hill: Delhi, 2002.

Course Code: ES 101
Course Title: Engineering Graphics

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
0	0	5	5	2.5	60	40	100	-	-	-
ET: End Term		PA: Progressive Assessment			Total marks (Th + Pr.) =100					

Course Content

Unit-I: Basic Elements of Drawing
Drawing Instruments and supporting materials: method to use them with applications. Convention of lines and their applications. Representative Fractions – reduced, enlarged and full size scales; Engineering
Scales such as plain and diagonal scale. Dimensioning techniques as per SP-46:2003 – types and applications of chain, parallel and coordinate dimensioning. Geometrical and Tangency constructions. (Redraw the figure)

Unit-II: Orthographic Projections
Introduction of projections-orthographic, perspective, isometric and oblique: concept and applications. (No question to be asked in examination). Introduction to orthographic projection, First angle and Third angle method, their symbols. Conversion of pictorial view into Orthographic Views – object containing plain surfaces, slanting surfaces, slots, ribs, cylindrical surfaces. (use First Angle Projection method only)

Unit-III: Isometric Projections
Introduction to isometric projections. Isometric scale and Natural scale. Isometric view and isometric projection. Illustrative problems related to objects containing lines, circles and arcs shape only. Conversion of orthographic views into isometric view/projection.

Unit-IV: Free Hand Sketches of Engineering Elements
Free hand sketches of machine elements: Thread profiles, nuts, bolts, studs, set screws, washer, Locking arrangements. (For branches other than mechanical Engineering, the teacher should select branch specific elements for free hand sketching) Free hand sketches of orthographic view (on squared graph paper) and isometric view (on isometric grid paper)

Unit-V: Computer Aided Drafting Interface
Computer Aided Drafting: concept. Hardware and various CAD software available. System requirements and Understanding the interface. Components of AutoCAD software window: Title bar, standard tool bar, menu bar, object properties tool bar, draw tool bar, modify tool bar, cursor cross hair. Command window, status bar, drawing area, UCS icon. File features: New file, Saving the file, Opening an existing drawing file, Creating templates, Quit. Setting up new drawing: Units, Limits, Grid, Snap. Undoing and redoing action.

Unit-VI: Computer Aided Drafting
Draw basic entities like Line, Circle, Arc, Polygon, Ellipse, Rectangle, Multiline, PolyLine. Method of Specifying points: Absolute coordinates, Relative Cartesian and Polar coordinates. Modify and edit commands like trim, extend, delete, copy, offset, array, block, layers. Dimensioning: Linear, Horizontal Vertical, Aligned, Rotated, Baseline, Continuous, Diameter, Radius, Angular Dimensions. Dim scale variable. Editing dimensions. Text: Single line Text, Multiline text. Standard sizes of sheet. Selecting Various plotting parameters such as Paper size, paper units, Drawing orientation, plot scale, plot offset, plot area, print preview.

Sr. No.	Practical Exercises	Unit	App. Hrs
1	Draw horizontal, Vertical, 30-degree, 45-degree, 60- and 75-degrees lines, different types of lines, dimensioning styles using Tee and Set squares/ drafter. (Do this exercise in sketch book)	I	02
2	Write alphabets and numerical (Vertical only) (Do this exercise in sketch book)	I	02
3	Draw regular geometric constructions and redraw the given figure (Do this exercise in sketch book) Part I	II	02
4	Draw regular geometric construction and redraw the given figure (Do this exercise in sketch book) Part II	II	02
5	Draw a problem on orthographic projections using first angle method of projection having plain surfaces and slanting. Part I	III	02
6	Draw another problem on orthographic projections using first angle method of projection having slanting surfaces with slots. Part II	III	02
7	Draw two problems on orthographic projections using first angle method of projection having cylindrical surfaces, ribs. Part I	III	02
8	Draw two problems on Isometric view of simple objects having plain and slanting surface by using natural scale. Part I	IV	02
9	Draw some problems on Isometric projection of simple objects having cylindrical surface by using isometric scale. Part I	IV	02
10	Draw free hand sketches/ conventional representation of machine elements in sketch book such as thread profiles, nuts, bolts, studs, set screws, washers, locking arrangements. Part I	V	02
11	Problem based Learning: Given the orthographic views of at least three objects with few missing lines, the student will try to imagine the corresponding objects, complete the views and draw these views in sketch book. Part I	II, III, V	02
12	Draw basic 2D entities like: Rectangle, Rhombus, Polygon using AutoCAD (Print out should be a part of progressive assessment). Part I	V	02
13	Draw basic 2D entities like: Circles, Arcs, circular using AutoCAD (Printout should be a part of progressive assessment). Part II	V	02
14	Draw basic 2D entities like: Circular and rectangular array using AutoCAD (Printout should be a part of progressive assessment). Part III	V	02
15	Draw blocks of 2D entities comprises of Rectangle, Rhombus, Polygon, Circles, Arcs, circular and rectangular array, blocks using AutoCAD (Print out should be a part of progressive assessment). Part IV	V	02
16	Draw basic branch specific components in 2D using AutoCAD (Print out should be a part of term work). Part I	VI	02
17	Draw complex branch specific components in 2D using AutoCAD (Print should be a part of progressive assessment). Part I	VI	02
		Total	**34**

Suggested Learning Resources:

1. Bureau of Indian Standards. *Engineering Drawing Practice for Schools and Colleges IS: Sp-46.* BIS. Government of India, Third Reprint, October 1998; ISBN: 81-7061-091-2.
2. Bhatt, N. D. *Engineering Drawing*. Charotar Publishing House, Anand, Gujrat 2010; ISBN: 978-93-80358-17-8.
3. Jain & Gautam, Engineering Graphics & Design, Khanna Publishing House, New Delhi (ISBN: 978-93-86173-478)
4. Jolhe, D. A. *Engineering Drawing*. Tata McGraw Hill Edu. New Delhi, 2010; ISBN: 978-0-07-064837-1
5. Dhawan, R. K. *Engineering Drawing*. S. Chand and Company, New Delhi; ISBN: 81-219-1431-0.

Course Code: ES 103
Course Title: Engineering Workshop Practice

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
0	0	3	3	1.5	-	-	-	40	60	100
ET: End Term		PA: Progressive Assessment			Total marks (Th + Pr.) =100					

Sr. No.	Details of Practial Content
I	**Carpentry:** i) Demonstration of different wood working tools / machines. ii) Demonstration of different wood working processes, like plaining, marking, chiseling, grooving, turning of wood etc. iii) One simple job involving any one joint like mortise and tenon dovetail, bridle, half lap etc.
II	**Fitting:** i) Demonstration of different fitting tools and drilling machines and power tools. ii) Demonstration of different operations like chipping, filing, drilling, tapping, sawing, cutting etc. iii) One simple fitting job involving practice of chipping, filing, drilling, tapping, cutting etc.
III	**Welding:** i) Demonstration of different welding tools / machines. ii) Demonstration on Arc Welding, Gas Welding, MIG, MAG welding, gas cutting and rebuilding of broken parts with welding. iii) One simple job involving butt and lap joint
IV	**Sheet Metal Working:** i) Demonstration of different sheet metal tools / machines. ii) Demonstration of different sheet metal operations like sheet cutting, bending, edging, end curling, lancing, soldering, brazing, and riveting. iii) One simple job involving sheet metal operations and soldering and riveting.
V	**Electrical House Wiring:** Practice on simple lamp circuits i) One lamp controlled by one switch by surface conduit wiring. ii) Lamp circuits- connection of lamp and socket by separate switches. iii) Connection of Fluorescent lamp/tube light. iv) Simple lamp circuits: Install bedroom lighting. v) Simple lamp circuits: Install stair case wiring.
VI	**Demonstration:** i) Demonstration of measurement of Current, Voltage, Power and Energy. ii) Demonstration of advance power tools, pneumatic tools, electrical wiring tools and accessories. iii) Tools for Cutting and drilling

Suggested Learning Resources:

1. S.K. Hajara Chaudhary, Workshop Technology, Media Promoters and Publishers, New Delhi, 2015
2. B.S. Raghuwanshi, Workshop Technology, Dhanpat Rai and sons, New Delhi 2014
3. K. Venkat Reddy, Workshop Practice Manual, BS Publications, Hyderabad 2014
4. Kents Mechanical Engineering Hand book, John Wiley and Sons, New York

Sample Path: SECOND Semester

Sl. No	Category of Course	Code No.	Course Title	Hours per week			Total contact hrs/ week	Credits
				L	T	P		
1	Basic Science	BS102	Mathematics-II	3	1	0	4	4
2		BS104	Applied Physics-II	2	1	2	5	4
3	Engineering Science	ES102	Introduction to IT Systems	3	0	4	7	5
4		ES104	Fundamentals of Electrical & Electronics Engineering	2	1	2	5	4
5		ES106	Engineering Mechanics	2	1	2	5	4
6	Audit	AU102	Environmental Science	2	0	0	2	0
						Total	**28**	**21**

Course Code: BS 102
Course Title: Mathematics-II

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
3	1	0	4	4	60	40	100	-	-	-
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.) = 100					

Course Content

Unit-I: Determinants and Matrices

Elementary properties of determinants up to 3rd order, consistency of equations, Crammer's rule.
Algebra of matrices, Inverse of a matrix, matrix inverse method to solve a system of linear equations in 3 variables.

Unit-II: Integral Calculus

Integration as inverse operation of differentiation.
Simple integration by substitution, by parts and by partial fractions (for linear factors only). Use of formulas
$\int_0^{\frac{\pi}{2}} sin^n x\, dx \int_0^{\frac{\pi}{2}} sin^n x\, dx$; $\int_0^{\frac{\pi}{2}} cos^n x\, dx \int_0^{\frac{\pi}{2}} cos^n x\, dx$ and
$\int_0^{\frac{\pi}{2}} sin^m x\, cos^n x\ \ dx \int_0^{\frac{\pi}{2}} sin^m x\, cos^n x\ \ dx$ for solving problems; where *m* and *n* are positive integers.
Applications of integration for
i. Simple problem on evaluation of area bounded by a curve and axes.
ii. Calculation of Volume of a solid formed by revolution of an area about axes. (Simple problems).

Unit – III: Co-Ordinate Geometry

Equation of straight line in various standard forms (without proof), intersection of two straight lines, angle between two lines. Parallel and perpendicular lines, perpendicular distance formula.
General equation of a circle and its characteristics.
To find the equation of a circle, given:

i) Centre and radius,
ii) Three points lying on it and
iii) Coordinates of end points of a diameter;

Definition of conics (Parabola, Ellipse, Hyperbola) their standard equations without proof. Problems on conics when their foci, directories or vertices are given.

Unit – IV: Vector Algebra

Definition notation and rectangular resolution of a vector. Addition and subtraction of vectors. Scalar and vector products of 2 vectors. Simple problems related to work, moment and angular velocity.

Unit-V: Differential Equations

Solution of first order and first degree differential equation by variable separation method (simple problems). PYTHON – Simple Introduction.

Suggested Learning Resources:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 40th Edition, 2007.
2. G. B. Thomas, R. L. Finney, Calculus and Analytic Geometry, Addison Wesley, 9th Edition, 1995.
3. S.S. Sabharwal, Sunita Jain, Eagle Parkashan, Applied Mathematics, Vol. I & II, Jalandhar.
4. Comprehensive Mathematics, Vol. I & II by Laxmi Publications, Delhi.
5. Reena Garg & Chandrika Prasad, Advanced Engineering Mathematics, Khanna Publishing House, New Delhi

Course Code: BS 104
Course Title: Applied Physics-II

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
2	1	2	5	4	60	40	100	40	60	100
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.) = 200					

Course Content

Unit-I: Wave motion and its applications

Wave motion, transverse and longitudinal waves with examples, definitions of wave velocity, frequency & wave length and their relationship, Sound & light waves and their properties, wave equation ($y = r sin\omega t$) amplitude, phase, phase difference, principle of superposition of waves and beat formation.
Simple Harmonic Motion (SHM): definition, expression for displacement, velocity, acceleration, time period, frequency etc. Simple harmonic progressive wave and energy transfer, study of vibration of cantilever and determination of its time period, Free, forced and resonant vibrations with examples.
Acoustics of buildings – reverberation, reverberation time, echo, noise, coefficient of absorption of sound, methods to control reverberation time and their applications.
Ultrasonic waves – Introduction and properties, engineering and medical applications of ultrasonic.

Unit- II: Optics

Basic optical laws; reflection and refraction, refractive index, Images and image formation by mirrors, lens and thin lenses, lens formula, power of lens, magnification and defects. Total internal reflection, Critical angle and conditions for total internal reflection, applications of total internal reflection in optical fiber.

Unit-III: Electrostatics

Coulombs law, unit of charge, Electric field, Electric lines of force and their properties, Electric flux, Electric potential and potential difference, Gauss law: Application of Gauss law to find electric field intensity of straight charged conductor, plane charged sheet and charged sphere.
Capacitor and its working, Types of capacitors, Capacitance and its units. Capacitance of a parallel plate capacitor, Series and parallel combination of capacitors (related numerical), dielectric and its effect on capacitance, dielectric break down.

Unit-IV: Current Electricity

Electric Current and its units, Direct and alternating current, Resistance and its units, Specific resistance, Conductance, Specific conductance, Series and parallel combination of resistances. Factors affecting resistance of a wire, carbon resistances and colour coding.
Ohm’s law and its verification, Kirchhoff’s laws, Wheatstone bridge and its applications (slide wire bridge only), Concept of terminal potential difference and Electro motive force (EMF)
Heating effect of current, Electric power, Electric energy and its units (related numerical problems), Advantages of Electric Energy over other forms of energy.

Unit-V: Electromagnetism

Types of magnetic materials; dia, para and ferromagnetic with their properties, Magnetic field and its units, magnetic intensity, magnetic lines of force, magnetic flux and units, magnetization.
Concept of electromagnetic induction, Faraday’s Laws, Lorentz force (force on moving charge in magnetic field). Force on current carrying conductor, force on rectangular coil placed in magnetic field.
Moving coil galvanometer; principle, construction and working, Conversion of a galvanometer into ammeter and voltmeter.

Unit-VI: Semiconductor Physics
Energy bands in solids, Types of materials (insulator, semi-conductor, conductor), intrinsic and extrinsic semiconductors, p-n junction, junction diode and V-I characteristics, types of junction diodes.
Diode as rectifier – half wave and full wave rectifier (centre taped).
Transistor; description and three terminals, Types- pnp and npn, some electronic applications (list only).
Photocells, Solar cells; working principle and engineering applications.

Unit-VII: Modern Physics
Lasers: Energy levels, ionization and excitation potentials; spontaneous and stimulated emission; population inversion, pumping methods, optical feedback, Types of lasers; Ruby, He-Ne and semiconductor, laser characteristics, engineering and medical applications of lasers. *(Short Discussion and Demonstration in Lab)*
Fiber Optics: Introduction to optical fibers, light propagation, acceptance angle and numerical aperture, fiber types, applications in; telecommunication, medical and sensors. *(Short Discussion and Demonstration in Lab with OFC)*

List of Practicals/Activities

1. To determine and verify the time period of a cantilever.
2. To determine velocity of ultrasonic in different liquids using ultrasonic interferometer.
3. To verify laws of reflection from a plane mirror/ interface.
4. To verify laws of refraction (Snell's law) using a glass slab.
5. To determine focal length and magnifying power of a convex lens.
6. To verify Ohm's law by plotting graph between current and potential difference.
7. To verify laws of resistances in series and parallel combination.
8. To find the frequency of AC main using electrical vibrator.
9. To verify Kirchhoff's law using electric circuits.
10. To study the dependence of capacitance of a parallel plate capacitor on various factors and determines permittivity of air at a place.
11. To find resistance of a galvanometer by half deflection method.
12. To convert a galvanometer into an ammeter.
13. To convert a galvanometer into a voltmeter.
14. To draw V-I characteristics of a semiconductor diode (Ge, Si) and determine its knee voltage.
15. To verify inverse square law of radiations using a photo-electric cell.
16. To measure wavelength of a He-Ne/diode laser using a diffraction grating.
17. To measure numerical aperture (NA) of an optical fiber.
18. Study of an optical projection system (OHP/LCD) - project report.

NB: ***From the list of Experiments, Four from the following may be kept as Demonstrative Experiments. In that situation, Minimum number of Eight practical experiments are to be performed by the students.***

2. To determine velocity of ultrasonic wave in different liquids using ultrasonic interferometer.
8. To find the frequency of AC main using electrical vibrator.
10. To study the dependence of capacitance of a parallel plate capacitor on various factors and determine permittivity of air at a place.
11. To find resistance of a galvanometer by half deflection method.
16. To measure wavelength of a He-Ne/diode laser using a diffraction grating.
17. To measure numerical aperture (NA) of an optical fiber.

Suggested Learning Resources:

1. Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi
2. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi
3. Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi
4. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi.
5. Modern approach to Applied Physics-I and II, AS Vasudeva, Modern Publishers.

Course Code: ES 102
Course Title: Introduction to IT Systems

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
3	0	4	7	5	60	40	100	40	60	100
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.) = 200					

Course Content

Unit-I: Basic Internet Skills
Understanding browser, efficient use of search engines, awareness about Digital India portals (state and national portals) and college portals.
General understanding of various computer hardware components – CPU, Memory, Display, Keyboard, Mouse, HDD and other Peripheral Devices.

Unit-II: OS Installation
Linux and MS Windows,
Unix Shell and Commands, vi editor.

Unit-III: HTML
HTML4, CSS, making basic personal webpage.

Unit-IV: Office Tools
OpenOffice Writer, OpenOffice Spreadsheet (Calc), OpenOffice Impress.

Unit-V: Information Security Best Practices
Class lectures will only introduce the topic or demonstrate the tool; actual learning will take place in Lab by practicing regularly.

List of Practical's/Activities

1. Browser features, browsing, using various search engines, writing search queries
2. Visit various e-governance/Digital India portals, understand their features, services offered
3. Read Wikipedia pages on computer hardware components, look at those components in lab, identify them, recognize various ports/interfaces and related cables, etc.
4. Install Linux and Windows operating system on identified lab machines, explore various options, do it multiple times
5. Connect various peripherals (printer, scanner, etc.) to computer, explore various features of peripheral and their device driver software.
6. Practice HTML commands, try them with various values, make your own Webpage
7. Explore features of Open Office tools, create documents using these features, do it multiple times
8. Explore security features of Operating Systems and Tools, try using them and see what happens.

Suggested Learning Resources:

1. R.S. Salaria, Computer Fundamentals, Khanna Publishing House
2. Ramesh Bangia, PC Software Made Easy – The PC Course Kit, Khanna Publishing House
3. Online Resources, Linux man pages, Wikipedia
4. Mastering Linux Shell Scripting: A practical guide to Linux command-line.
5. Bash scripting and Shell programming, by Mokhtar Ebrahim, Andrew Mallett

Course Code: ES 104
Course Title: Fundamentals of Electrical & Electronics Engineering

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
2	1	2	5	4	60	40	100	40	60	100
ET: End Term		PA: Progressive Assessment			Total Marks (Th + Pr.): 200					

Course Content

Unit-I: Overview of Electronic Components & Signals
Passive Active Components: Resistances, Capacitors, Inductors, Diodes, Transistors, FET, MOS and CMOS and their Applications.
Signals: DC/AC, voltage/current, periodic/non-periodic signals, average, rms, peak values, different types of signal waveforms, Ideal/non-ideal voltage/current sources, independent/ dependent voltage current sources.

Unit-II: Overview of Analog Circuits
Operational Amplifiers-Ideal Op-Amp, Practical op amp, Open loop and closed loop configurations,
Application of Op-Amp as amplifier, adder, differentiator and integrator.

Unit-III: Overview of Digital Electronics
Introduction to Boolean Algebra, Electronic Implementation of Boolean Operations, Gates-Functional Block Approach, Storage elements-Flip Flops-A Functional block approach, Counters: Ripple, Up/down and decade, Introduction to digital IC Gates (of TTL Type).

Unit-IV: Electric and Magnetic Circuits
EMF, Current, Potential Difference, Power and Energy; M.M.F, magnetic force, permeability, hysteresis loop, reluctance, leakage factor and BH curve; Electromagnetic induction, Faraday's laws of electromagnetic induction, Lenz's law; Dynamically induced emf; Statically induced emf; Equations of self and mutual inductance; Analogy between electric and magnetic circuits.

Unit-V: A.C. Circuits
Cycle, Frequency, Periodic time, Amplitude, Angular velocity, RMS value, Average value, Form Factor Peak Factor, impedance, phase angle, and power factor; Mathematical and phasor representation of alternating emf and current; Voltage and Current relationship in Star and Delta connections; A.C in resistors, inductors and capacitors; A.C in R-L series, R-C series, R-L-C series and parallel circuits; Power in A. C. Circuits, power triangle.

Unit-VI: Transformer and Machines
General construction and principle of different type of transformers; Emf equation and transformation ratio of transformers; Auto transformers; Construction and Working principle of motors; Basic equations and characteristic of motors.

List of Practical's/Activities

1. Determine the permeability of magnetic material by plotting its B-H curve.
2. Measure voltage, current and power in 1-phase circuit with resistive load.
3. Measure voltage, current and power in R-L series circuit.
4. Determine the transformation ratio (K) of 1-phase transformer.
5. Connect single phase transformer and measure input and output quantities.
6. Make Star and Delta connection in induction motor starters and measure the line and phase values.
7. Identify various passive electronic components in the given circuit

8. Connect resistors in series and parallel combination on bread board and measure its value using digital multimeter.
9. Connect capacitors in series and parallel combination on bread board andmeasure its value using multimeter.
10. Identify various active electronic components in the given circuit.
11. Use multimeter to measure the value of given resistor.
12. Use LCR-Q tester to measure the value of given capacitor and inductor.
13. Determine the value of given resistor using digital multimeter to confirm with colour code.
14. Test the PN-junction diodes using digital multimeter.
15. Test the performance of PN-junction diode.
16. Test the performance of Zener diode.
17. Test the performance of LED.
18. Identify three terminals of a transistor using digital multimeter.
19. Test the performance of NPN transistor.
20. Determine the current gain of CE transistor configuration.
21. Test the performance of transistor switch circuit.
22. Test the performance of transistor amplifier circuit.
23. Test Op-Amp as amplifier and Integrator

Suggested Learning Resources:

1. Ritu Sahdev, Basic Electrical Engineering, Khanna Publishing House
2. Mittle and Mittal, Basic Electrical Engineering, McGraw Education, New Delhi, 2015
3. Saxena, S. B. Lal, Fundamentals of Electrical Engineering, Cambridge University Press
4. Theraja, B. L., Electrical Technology Vol – I, S. Chand Publications, New Delhi, 2015
5. Theraja, B. L., Electrical Technology Vol – II, S. Chand Publications, New Delhi, 2015

Course Code: ES 106
Course Title: Engineering Mechanics

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
2	1	1	5	4	60	40	100	40	60	100
ET: End Term		PA: Progressive Assessment			Total marks (Th + Pr.) = 200					

Course Content

Unit-I: Basics of Mechanics and Force System

Significance and relevance of Mechanics, Applied mechanics, Statics, Dynamics. Space, time, mass, particle, flexible body and rigid body. Scalar and vector quantity, Units of measurement (SI units) - Fundamental units and derived units.
Force – unit, representation as a vector and by Bow's notation, characteristics and effects of a force, Principle of transmissibility of force, Force system and its classification.
Resolution of a force - Orthogonal components of a force, moment of a force, Varignon's Theorem.
Composition of forces – Resultant, analytical method for determination of resultant for concurrent, non-concurrent and parallel co-planar force systems – Law of triangle, parallelogram and polygon of forces.

Unit-II: Equilibrium

Equilibrium and Equilibrant, Free body and Free body diagram, Analytical and graphical methods of analysing equilibrium
Lami's Theorem – statement and explanation, Application for various engineering problems.
Types of beam, supports (simple, hinged, roller and fixed) and loads acting on beam (vertical and inclined point load, uniformly distributed load, couple),
Beam reaction for cantilever, simply supported beam with or without overhang – subjected to combination of Point load and uniformly distributed load.
Beam reaction graphically for simply supported beam subjected to vertical point loads only.

Unit-III: Friction

Friction and its relevance in engineering, types and laws of friction, limiting equilibrium, limiting friction, co-efficient of friction, angle of friction, angle of repose, relation between co-efficient of friction and angle of friction.
Equilibrium of bodies on level surface subjected to force parallel and inclined to plane.
Equilibrium of bodies on inclined plane subjected to force parallel to the plane only.

Unit-IV: Centroid and Centre of Gravity

Centroid of geometrical plane figures (square, rectangle, triangle, circle, semi-circle, quarter circle)
Centroid of composite figures composed of not more than three geometrical figures
Centre of Gravity of simple solids (Cube, cuboid, cone, cylinder, sphere, hemisphere) Centre of
Gravity of composite solids composed of not more than two simple solids.

Unit-V: Simple Lifting Machine

Simple lifting machine, load, effort, mechanical advantage, applications and advantages. Velocity ratio, efficiency of machines, law of machine.
Ideal machine, friction in machine, maximum Mechanical advantage and efficiency, reversible and non-reversible machines, conditions for reversibility
Velocity ratios of Simple axle and wheel, Differential axle and wheel, Worm and worm wheel,
Single purchase and double purchase crab winch, Simple screw jack, Weston's differential pulley block, geared pulley block.

List of Practical's/Activities

1. To study various equipment's related to Engineering Mechanics.
2. To find the M.A., V.R., Efficiency and law of machine for Differential Axle and Wheel.
3. To find the M.A., V.R., Efficiency and law of machine for Simple Screw Jack.
4. Derive Law of machine using Worm and worm wheel.
5. Derive Law of machine using Single purchase crab.
6. Derive Law of machine using double purchase crab.
7. Derive Law of machine using Weston's differential or wormed geared pulley block.
8. Determine resultant of concurrent force system applying Law of Polygon of forces using force table.
9. Determine resultant of concurrent force system graphically.
10. Determine resultant of parallel force system graphically.
11. Verify Lami's theorem.
12. Study forces in various members of Jib crane.
13. Determine support reactions for simply supported beam.
14. Obtain support reactions of beam using graphical method.
15. Determine coefficient of friction for motion on horizontal and inclined plane.
16. Determine centroid of geometrical plane figures.

Suggested Learning Resources:

1. D.S. Bedi, Engineering Mechanics, Khanna Publications, New Delhi (2008)
2. Khurmi, R.S., Applied Mechanics, S. Chand & Co. New Delhi.
3. Bansal R K, A text book of Engineering Mechanics, Laxmi Publications.
4. Ramamrutham, Engineering Mechanics, S. Chand & Co. New Delhi.
5. Dhade, Jamadar & Walawelkar, Fundamental of Applied Mechanics, Pune Vidhyarthi Gruh.

Course Code: AU 102
Course Title: Environmental Science

Contact Hours per Week					Evaluation Scheme					
Lecture	Tutorial	Practical	Total	Credits	Theory			Practical		
(L)	(T)	(P)	L+T+P	(C)	ET	PA	Total	ET	PA	Total
2	0	0	2	0	-	-	-	-	-	-
ET: End Term		PA: Progressive Assessment			Total marks (Th + Pr.) = Not Applicable					

Course Content

Unit-I: Ecosystem
Structure of ecosystem, Biotic & Abiotic components;
Food chain and food web
Aquatic (Lentic and Lotic) and terrestrial ecosystem
Carbon, Nitrogen, Sulphur, Phosphorus cycle.
Global warming -Causes, effects, process, Green House Effect, Ozone depletion

Unit-II: Air and Noise Pollution
Definition of pollution and pollutant, Natural & manmade sources of air pollution (Refrigerants, I.C., Boiler)
Air Pollutants: Types, Particulate Pollutants: Effects and control
(Bag filter, Cyclone separator, Electrostatic Precipitator)
Gaseous Pollution Control: Absorber, Catalytic Converter, Effects of air pollution due to Refrigerants, I.C., Boiler
Noise pollution: sources of pollution, measurement of pollution level, Effects of Noise pollution, Noise pollution (Regulation and Control) Rules, 2000

Unit-III: Water and Soil Pollution
Sources of water pollution, Types of water pollutants, Characteristics of water pollutants Turbidity, pH, total suspended solids, total solids BOD and COD: Definition, calculation
Waste Water Treatment: Primary methods: sedimentation, froth floatation, Secondary methods: Activated sludge treatment, RO (reverse osmosis).
Causes, Effects and Preventive measures of Soil Pollution: Causes-Excessive use of Fertilizers, E-Waste.

Unit-IV: Renewable sources of Energy
Solar Energy: Basics of Solar energy. Flat plate collector (Liquid & Air).
Biomass: Overview of biomass as energy source. Utilization and storage of biogas.
Wind energy: Current status and future prospects of wind energy. Wind energy in India. Environmental benefits and problem of wind energy.
New Energy Sources: Need of new sources. Different types new energy sources. Applications of (Hydrogen energy, Ocean energy resources, Tidal energy conversion.) Concept, origin and power plants of geothermal energy

Unit-V: Solid Waste Management, ISO 14000 & Environmental Management
Solid waste generation- Sources and characteristics and Management.
Collection and disposal: MSW (3R, principles, energy recovery, sanitary landfill), Hazardous waste
Structure and role of Central and state pollution control board.
Concept of Carbon Credit, Carbon Footprint. Environmental management in fabrication industry.
ISO14000: Introduction and Benefits.

Suggested Learning Resources:
1. S.C. Sharma & M.P. Poonia, Environmental Studies, Khanna Publishing House, New Delhi
2. C.N. R. Rao, Understanding Chemistry, Universities Press (India) Pvt. Ltd., 2011.
3. Nazaroff, William, Cohen, Lisa, Environmental Engineering Science, Willy, New York, 2000.
4. O.P. Gupta, Elements of Environmental Pollution Control, Khanna Publishing House, New Delhi
5. Rao, M. N. Rao, H.V.N, Air Pollution, Tata Mc-Graw Hill Publication, New Delhi, 1988.